NetFRAME

Network Topology & Systems Reference

`192.168.10.0/24 · km-cluster · 7 nodes · 3 GPUs · RKE2`

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Document	NetFRAME Network Topology & Systems Reference
Version	2.0
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Classification	Public / Sanitized (public IPs, service tags, serials, and secrets omitted)
Method	Live discovery (Proxmox API, guest-agent, `kubectl`, ZFS, SSH facts) reconciled against the Home-Lab vault
Scope	Every host, VLAN, subnet, WAN, link, service, pool, cluster, and overlay path

Single-site production-grade home lab · Proxmox VE 9.2.3 · Juniper-cored VLAN fabric · ZFS + PBS · RKE2 Kubernetes · private GPU LLM platform

Contents

1 Executive Summary

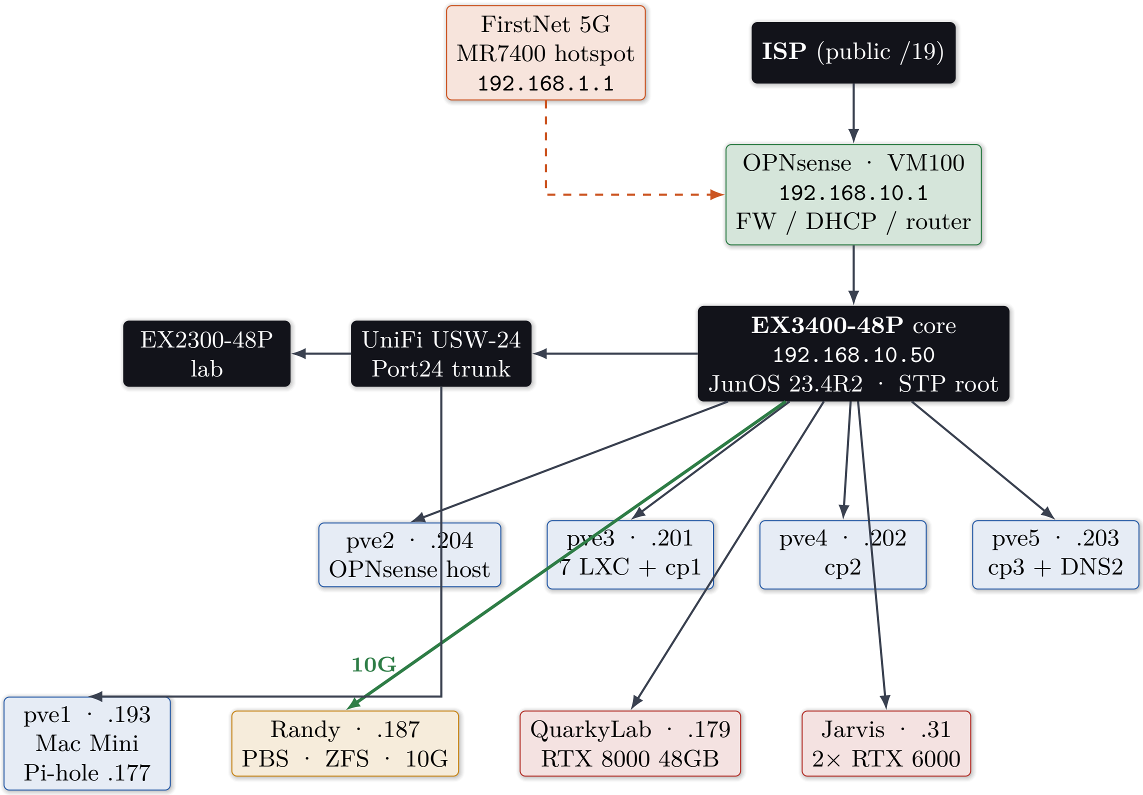
NetFRAME is a single-site, 42U home lab built around a **7-node Proxmox VE 9.2.3 cluster** (`km-cluster`), a **Juniper-cored, VLAN-segmented fabric**, and a **24-bay ZFS storage node** providing NFS and Proxmox Backup Server. It runs reverse-proxied web services with automated TLS, a Prometheus / Grafana / Loki observability stack with Discord alerting, drive-health telemetry across roughly 50 disks, a Wazuh SIEM, an internal step-ca PKI, a password vault, a **3-node HA RKE2 Kubernetes cluster**, and a **GPU-accelerated private LLM platform** (RTX 8000 plus dual RTX 6000, roughly 96 GB VRAM) with a custom OpenAI-compatible router that transparently escalates to the Claude API. The site rides **dual WAN** (ISP primary, FirstNet 5G backup) and **HA DNS**.

Proxmox nodes (quorate)	7/7	ZFS raw across pools	~95 TB
Standalone hypervisor	1 (pve1)	PBS datastore	24.6 TB
Cluster VMs / LXCs	5 / 7	Monitored drives	~50
Managed switches	3	Overlay VPN nodes	6
VLANs defined	7	Aggregate RAM	~1.2 TB
Physical GPUs / VRAM	3 / ~96 GB	Kubernetes nodes	4 (3 CP + 1 agent)
WAN uplinks	2 (ISP + 5G)	UPS (split-bus)	2 / ~2220 W

Edge model: single-NAT with dual-WAN failover

OPNsense (VM 100 on `pve2`) is the network edge. Its primary WAN sits *directly* on the ISP public assignment (DHCP, a routable `/19`), so the lab is single-NAT, not double-NAT. A second WAN interface carries a **FirstNet 5G hotspot** (`192.168.1.0/24`) as an automatic backup uplink via an OPNsense gateway group. OPNsense is also the LAN router / firewall / DHCP for `192.168.10.0/24` and the inter-VLAN gateway. (v1.0 of this document described a UniFi Dream Router double-NAT edge; that was incorrect and is corrected here.)

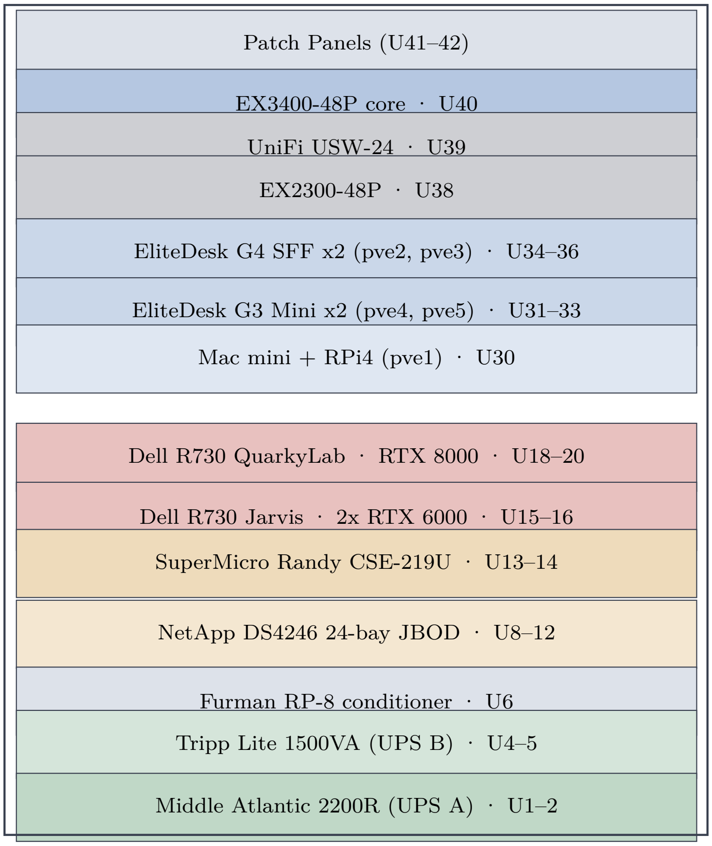
2 Physical & Logical Topology



Solid green = 10 GbE (Randy `xe-0/2/0`; Jarvis `xe-0/2/2`). Dashed orange = 5G backup WAN into OPNsense `nic2`. All other links 1 GbE.

3 Layer 1: Physical

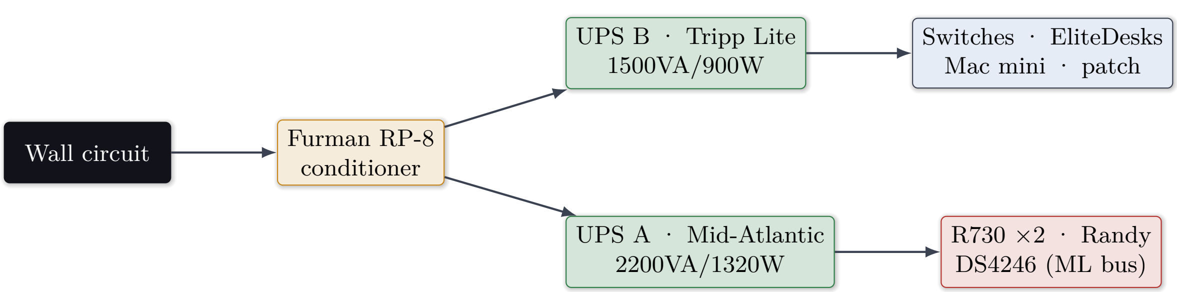
3.1 Rack elevation (NetFRAME CS9000, 42U, rear panel removed for R730 depth)



3.2 Key physical links

From	Port	To	Speed	Notes
EX3400	`ge-0/0/46`	UniFi USW-24 P24	1G	802.1Q trunk, native 1 + tagged 20/30/40/50/60/70
EX3400	`ge-0/0/45`	EX2300	1G	Trunk, all VLANs
EX3400	`xe-0/2/0`	Randy nic3 (Mellanox)	**10G**	Storage / NFS / PBS data path
EX3400	`xe-0/2/2`	Jarvis ConnectX	**10G**	Access VLAN 30
EX3400	`ge-0/0/24`	QuarkyLab nic2	1G	Trunk native 1 + tagged 30
EX3400	`ge-0/0/22`	Jarvis nic2	1G	Native VLAN 1 (mgmt / corosync)
EX3400	`ge-0/0/30,32,44`	iDRAC / IPMI OOB	1G	Tagged VLAN 20 only
pve2	`nic0` onboard	ISP handoff	1G	Primary WAN, vtnet0, public /19
pve2	`nic2` (I350)	MR7400 5G hotspot	2.5G	Backup WAN2, vtnet2, direct cable
EX3400	`xe-0/2/3`	UniFi SFP2 DAC	–	**DOWN**, 10G/1G EEPROM mismatch

3.3 Power: split-bus one-line



Capacity watch

UPS A carries both GPU R730s. The RTX 8000 alone can draw roughly 260 W under CUDA; with Jarvis's 2× RTX 6000 online, full ML load approaches the 1320 W continuous rating. Both UPS units are exposed via **NUT 2.8.1** on `pve3:3493` (UPS A over SNMP card `192.168.10.180`, UPS B over USB).

4 Layer 2: Switching & VLANs

4.1 Switch fabric

Device	Mgmt IP	OS	Role
EX3400-48P	`192.168.10.50`	JunOS 23.4R2-S7.4	**Core**: 48×1G PoE+, 4×10G SFP+, 2×40G QSFP+, dual PSU, STP root (prio 4096)
UniFi USW-24-250W	UniFi-managed	UniFi	Access / AP aggregation; Port 24 trunk
EX2300-48P	via trunk	JunOS	Secondary / lab isolation

4.2 VLAN matrix

VLAN 1	`192.168.10.0/24`	Management
VLAN 20	`192.168.20.0/24`	Trusted/OOB
VLAN 30	`192.168.30.0/24`	Servers
VLAN 40	`192.168.40.0/24`	IoT
VLAN 50	`192.168.50.0/24`	VoIP
VLAN 60	`192.168.60.0/24`	Guest
VLAN 70	`192.168.70.0/24`	Lab

Gateways are OPNsense SVIs (`192.168.X.1`); the EX3400 carries the L3 mgmt SVI on `irb.10`.

JunOS ELS gotcha (live)

On this EX3400, `native-vlan-id` must be set at the **physical-interface** level, *not* under `unit 0 family ethernet-switching`. Misplacement previously caused a trunk outage. The tagged trunk to UniFi is `ge-0/0/46`.

5 Layer 3: Addressing, Routing, DNS

5.1 WAN uplinks

Uplink	OPNsense iface	Addressing	Role
WAN (primary)	vtnet0 (nic0)	ISP DHCP, public /19	Default route; direct ISP handoff (single-NAT)
WAN2 (backup)	vtnet2 (nic2)	`192.168.1.0/24` DHCP	FirstNet 5G MR7400 hotspot; gateway-group failover

The public WAN address is withheld from this public edition. WAN2 failover is monitor-driven (dpinger against a far target), with LAN and per-VLAN internet rules pointed at the gateway group.

5.2 Static IP registry (VLAN 1)

IP	Host	Role
`.1`	OPNsense	LAN gateway / FW / DHCP / dual-WAN (VM100, pve2)
`.2`	UniFi USW-24	Access switch mgmt
`.31`	Jarvis	R730 LLM node
`.50`	EX3400	Core switch
`.51-.53`	rke2-cp1/2/3	RKE2 control-plane VMs (201/202/203)
`.54`	RKE2 API VIP	kube-vip virtual IP (:6443)
`.71-.75`	MetalLB pool	Kubernetes LoadBalancer range (registry on `.72`)
`.100/.199`	Ares	Admin workstation (wired / WiFi)
`.148`	Homepage	LXC 106
`.177`	Pi-hole (primary)	LXC 103 on pve1 (Mac Mini)
`.178`	Pi-hole (secondary)	CT 108 on pve5 (DNS HA mirror)
`.179`	QuarkyLab	R730 ML node
`.180`	UPS A SNMP	Mid-Atlantic OL2200R card
`.181`	Nginx Proxy Mgr	LXC 101
`.182`	Vaultwarden	LXC 102
`.183`	Grafana / Prom / Loki / Influx / Scrutiny	LXC 103
`.184`	Wazuh SIEM	VM 104 (QuarkyLab)
`.185`	Open WebUI	LXC 107
`.186`	Headscale	LXC 105
`.187`	Randy	PBS / ZFS / Jellyfin / RKE2 agent
`.193`	pve1	Mac Mini standalone (Pi-hole host)
`.201-.204`	pve3 / pve4 / pve5 / pve2	Cluster nodes

5.3 Routing: per-node default gateway (live 2026-07-13)

Node	Default GW	Iface	Note
pve2	`192.168.10.1`	vmbr1	correct (OPNsense)
pve3 / 4 / 5	`192.168.10.1`	vmbr0	corrected 2026-07-13 (was `192.168.1.1`, F-1 resolved)
QuarkyLab	`192.168.30.1`	vmbr0.30	VLAN 30 egress
Jarvis	`192.168.30.1`	vmbr1	VLAN 30 egress (10G)
Randy	`192.168.30.1`	vmbr0.30	VLAN 30 egress
rke2-cp1/2/3	`192.168.10.1`	ens18	verified via hostNetwork pods

5.4 DNS (HA) & OOB / Servers VLANs

DNS is highly available. The primary Pi-hole (`192.168.10.177`, LXC on pve1) and a secondary (`192.168.10.178`, CT 108 on pve5) both resolve `*.netframe.local` (e.g. `llm.` / `chat.` → NPM `.181`) and sink ads/trackers. OPNsense DHCP hands out *both* resolvers on all 7 VLAN scopes, so clients fail over automatically; the secondary is a nebula-sync full mirror of the primary (gravity, adlists, local DNS) refreshed every 15 minutes. Public `*.kylemason.org` resolves via Cloudflare (grey-cloud) to NPM.

IP	Host	Note
`192.168.20.20/21/22`	QuarkyLab iDRAC / Jarvis iDRAC / Randy IPMI	OOB VLAN 20 (2026-07-03); credentials rotated
`192.168.30.179/187/31`	QuarkyLab / Randy / Jarvis	NFS / PBS / egress (dual-homed)

6 Overlay: Headscale Mesh VPN

Self-hosted Headscale v0.29.1 (LXC 105, `192.168.10.186:8080`), tailnet `100.64.0.0/10`, replacing the Tailscale SaaS control plane.

Tailnet IP	Node	Tailnet IP	Node
`100.64.0.1`	Ares	`100.64.0.2`	Randy
`100.64.0.3`	pve5	`100.64.0.4`	pve4
`100.64.0.5`	pve3	`100.64.0.6`	Jarvis

7 Compute & Virtualization: km-cluster

7.1 Node hardware inventory

Node	Chassis	CPU	RAM	GPU	Kernel
pve2	EliteDesk 800 G4	i7-8700	32G	–	7.0.12-pve
pve3	EliteDesk 800 G4	i7-8700	48G	–	7.0.12-pve
pve4	EliteDesk 800 G3 Mini	i5-7500T	32G	–	7.0.12-pve
pve5	EliteDesk 800 G3 Mini	i5-7500T	32G	–	7.0.12-pve
QuarkyLab	Dell R730	2× E5-2699v4	512G	RTX 8000 48G	**6.14.11-pve**
Jarvis	Dell R730	2× E5-2687Wv4	384G	2× RTX 6000	**6.14.11-pve**
Randy	SuperMicro CSE-219U	2× E5-2690v3	128G	RX 580	7.0.12-pve
pve1	Mac Mini (standalone)	–	–	–	Pi-hole host

Kernel pinning: do not upgrade

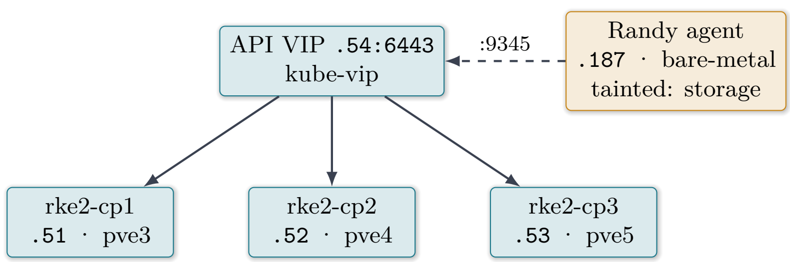
QuarkyLab and Jarvis are pinned to **6.14.11-9-pve** via `GRUB_DEFAULT`; 6.17+ breaks the NVIDIA 550.163.01 stack. Never run kernel upgrades or change the GRUB default on either GPU node.

7.2 VM / LXC inventory

ID	Name	Host	Res	IP	Payload
VM100	OPNsense	pve2	4G	`.1`	LAN router / FW / DHCP / dual-WAN v25.7
VM104	Wazuh	QuarkyLab	16G	`.184`	SIEM (guest-agent installed)
VM201	rke2-cp1	pve3	–	`.51`	RKE2 control plane / etcd
VM202	rke2-cp2	pve4	–	`.52`	RKE2 control plane / etcd
VM203	rke2-cp3	pve5	–	`.53`	RKE2 control plane / etcd
101	nginx-proxy	pve3	512M/8G	`.181`	Nginx Proxy Manager
102	vaultwarden	pve3	512M/10G	`.182`	Vaultwarden
103	grafana	pve3	2G/20G	`.183`	Grafana + Prom + Loki + Influx + Scrutiny
105	headscale	pve3	512M/4G	`.186`	Mesh VPN control
106	homepage	pve3	512M/8G	`.148`	Homepage + PeaNUT
107	openwebui	pve3	4G/16G	`.185`	Open WebUI
108	netframe-pihole2	pve5	–	`.178`	Secondary Pi-hole (DNS HA)

8 Kubernetes: RKE2 Platform

A 3-node HA RKE2 cluster (Phases 1–7 complete, 2026-07-10/11) overlays the Proxmox estate without disturbing existing workloads.



- **Control plane:** 3 VMs (201/202/203 on pve3/4/5), etcd quorum (2 of 3), Cilium CNI, kube-vip API VIP `192.168.10.54:6443`. CP nodes are schedulable (no dedicated workers, to avoid GPU-server contention).
- **Storage worker:** Randy joined as a bare-metal agent on the PVE storage host, tainted `role=storage:NoSchedule` with kubelet reservations so ZFS/PBS are not cgroup-throttled. Agents register on the supervisor port `:9345`, not the apiserver.
- **Networking / storage:** MetalLB pool `.71-.75`; Randy `datastore/k8s` NFS is the default StorageClass (`nfs-client`).
- **Registry:** private image registry on Randy local-ZFS (`registry.netframe.local`, MetalLB `.72`), step-ca TLS with 8-hour auto-renew, trusted by every node's containerd. GPU scheduling is deferred (SLURM and Ollama own the cards).

9 Storage Architecture

9.1 Randy: two ZFS pools

Pool	Layout	Raw	Usable	Notes
`datastore`	4× RAIDZ2 (3× 6-wide Toshiba + 1× 4-wide Seagate)	36.7 TB	~23 TB	PBS + primary NFS; ONLINE
`bulk`	2× 8-wide RAIDZ2 (DS4246, 16× 4TB SAS)	58.2 TB	~41.3 TiB	Built 2026-07-08, reboot-verified; ONLINE

Boot: RAID-1 mirror (2× ST200FM0053 SAS) on **AVAGO 3108 MegaRAID**. `datastore` data drives hang off a **separate LSI 9207-8e in IT mode**; the DS4246 shelf reaches Randy through a **second LSI 9207-8e (IT mode) with dm-multipath** across the shelf's two I/O modules. PBS v4.2.2 (`192.168.10.187:8007`) exposes a 24.6 TB datastore.

DS4246 buildout complete

The NetApp DS4246 shelf's dual-path SAS presented each disk on two paths (duplicate serial pairs). Multipath was configured before pooling, and pool `bulk` (2× 8-wide RAIDZ2, ~41.3 TiB usable) is now built and auto-imports cleanly across reboots. (v1.0 listed this as "buildout in progress".)

9.2 Jarvis: local ZFS

On the onboard LSI SAS-3 3008 (HBA330, IT mode): `tank` = raidz1 5× 2TB (**7.2T**, model library and bulk), `scratch` = 1× 200GB SSD (fast scratch). Ollama's `OLLAMA_MODELS` lives on `tank/models`.

PBS storage pin (F-4, resolved)

The VLAN-30 migration silently repointed PBS storage to `.30.187` and stalled VLAN-1 node backups from 2026-07-02; fixed 2026-07-06 back to `192.168.10.187`. A 10G-only `randy-pbs-10g` target on `.30.187` serves the GPU nodes. **Keep the general PBS storage on `.10.187`.**

10 Services & Application Layer

10.1 Reverse proxy & published services (NPM, CT101)

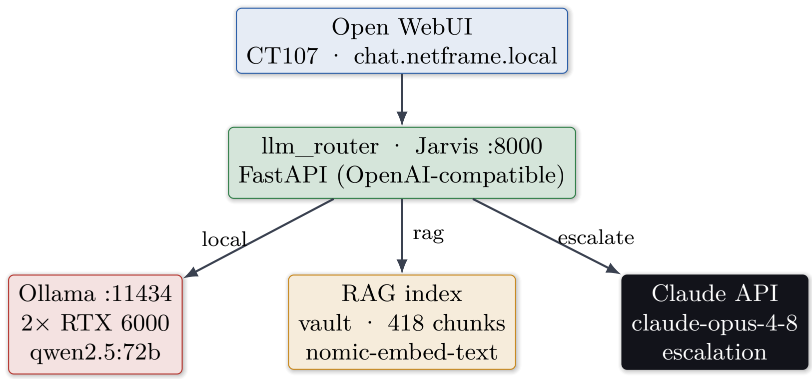
Hostname	Backend	TLS	Auth
vault.kylemason.org	`.182`	CF DNS-01	Vaultwarden
grafana.kylemason.org	`.183:3000`	CF DNS-01	Grafana
homepage.kylemason.org	`.148:3000`	CF DNS-01	Basic (kyle)
wazuh.kylemason.org	`.184:443`	CF DNS-01	Wazuh
health.kylemason.org	NetFRAME monitor	CF DNS-01	Basic (auth)
chat.netframe.local	Open WebUI `.185`	internal	admin
llm.netframe.local	Jarvis `.31:8000`	HTTP	Pi-hole-scoped

Observability: Prometheus (`.183:9090`, localhost-only, 8 exporters), Grafana v13, Loki, InfluxDB, Scrutiny (~50 drives, collectors on Randy/QuarkyLab/Jarvis), NUT (`pve3:3493`), CrowdSec IPS, step-ca PKI (`pve2:443`).

Alerting: Grafana to Discord (no Alertmanager)

Grafana Unified Alerting posts to two Discord channels (infra + UPS). Rules cover instance-down, both Pi-holes, ZFS pool degraded, GPU temp/memory, disk/memory pressure, registry cert expiry, a backup-verify dead-man's switch, and **cluster quorum** (`node_pve_cluster_quorate` textfile collector on all 7 nodes). Config-as-code lives in a private repo; a nightly Ansible `backup_verify` role checks per-guest PBS coverage.

11 AI / LLM Platform



On-prem, cited answers over your own notes

The `"rag"` model grounds answers on the Home-Lab Obsidian vault (418 chunks, cosine index) with `[source]` citations, served entirely on GPU on Jarvis, with a one-flag escalation to Claude Opus for hard queries. QuarkyLab's RTX 8000 48 GB drives a parallel DUNE research RAG agent. Ollama model weights live on the `tank/models` ZFS dataset.

12 Security Architecture

- **OOB isolation (Phase 1):** all three BMCs moved off flat VLAN 1 onto tagged VLAN 20; default credentials rotated to Vaultwarden; reachable only from Ares `enp0s31f6.20`.
- **Services segmentation (Phases 2–3):** Vaultwarden and Open WebUI on VLAN 30; a VLAN30-to-VLAN1 management clamp; DNS pass-rules verified pass-before-block.
- **Servers VLAN 30:** GPU / storage nodes dual-homed (NFS/PBS/egress on 30; corosync/mgmt on 1).
- **Ingress chokepoint:** all web traffic via NPM; admin planes localhost- or Ares-bound (pentest F-03/F-05, verified).
- **Resilience:** OPNsense serial-console DR verified, nightly age-encrypted `config.xml` backup to a private repo; guest-agent graceful shutdown on OPNsense and Wazuh; CI/CD lint gates on four repos.
- **Defense in depth:** Wazuh SIEM + CrowdSec IPS + step-ca PKI + Vaultwarden secrets.

13 Data-Flow Walkthroughs

A · Grounded chat: Browser → Pi-hole resolves `chat.netframe.local` → NPM `.181` → Open WebUI `.185` → `llm_router` Jarvis `.31:8000` (`rag`) → embed + cosine search vault → Ollama `qwen2.5:72b` tensor-split → cited answer.

B · Nightly backup: `pvescheduler` → `vzdump` (02:00 CT / 03:00 VM) → PBS `192.168.10.187:8007` (VLAN 1) → prune 7d+4w.

C · Storage: GPU nodes mount Randy NFS over **VLAN 30 10G** (`.30.187`, `xe-0/2/0`); corosync stays on VLAN 1.

D · WAN failover: ISP link drops → dpinger marks the primary gateway down → OPNsense gateway group shifts default egress to WAN2 (FirstNet 5G) → LAN and per-VLAN rules follow the group → fail-back when the ISP monitor recovers.

E · BMC access: Ares tags VLAN 20 (`enp0s31f6.20`) → EX3400 `ge-0/0/30/32/44` → iDRAC `192.168.20.20-22`.

14 Observations, Drift & Recommendations

#	State	Finding	Resolution / recommendation
F-1	✓	pve3/4/5 default GW was `192.168.1.1` (off-subnet, stale)	Repointed to `192.168.10.1` on 2026-07-13
F-3	✓	DS4246 dual-path duplicates; pool not built	Multipath configured; pool **`bulk`** built and reboot-verified
F-4	✓	PBS storage repointed to `.30.187` (broke backups)	Fixed; general storage on `.10.187`; 10G target split out
F-5	✓	Wazuh VM lacked guest-agent	Installed 2026-07-10; graceful shutdown confirmed
F-2	Low	Ares wired mgmt leg has flapped; VLAN 20 reroutes via WiFi	Treat wired as authoritative; verify **`ip route get`**
F-6	Low	`xe-0/2/3` DAC down (EEPROM mismatch)	Replace SFP or decommission
F-7	Info	VLANs 40/50/60/70 lightly populated	Populate or prune
F-8	Low	Amazon Echo on flat mgmt VLAN 1; benign non-relaying SOCKS5 artifact	Move to IoT VLAN 40 (egress-only)
F-9	WIP	RKE2 control-plane VMs lack **`qemu-guest-agent`**	Install pinned (channel enabled; package pending host access)
F-10	WIP	WAN2 gateway-group failover mid-deployment	Finish monitor IP + controlled failover test

15 Failure-Domain / Blast-Radius

Component	If it fails	Mitigation
pve2 (OPNsense)	No LAN routing / DHCP / inter-VLAN, whole LAN	onboot; serial-console + age-encrypted config DR
ISP uplink	Internet egress	FirstNet 5G WAN2 gateway-group failover
pve3 (7 LXCs + cp1)	Core services + one etcd member down	PBS backups; etcd 2-of-3; migratable CTs
Randy	Backups + GPU NFS + Jellyfin + k8s storage	RAIDZ2 (2-disk/vdev); split HBAs; multipath
EX3400	Entire wired fabric	Dual PSU; UPS B
UPS A	Both R730s + Randy + DS4246	2200VA headroom; graceful shutdown
DNS	Name resolution / ad-block	Two Pi-holes (`.177`/`.178`), both in DHCP
Corosync	Loss of quorum	7 votes on stable VLAN-1 L2; quorum alerting

16 Appendix: Live Port / Service Scan

Verified 2026-07-08 (**`nmap -sT -sV`**, top-200); RKE2 nodes added 2026-07-13. `:9100`=node-exporter, `:3128`=Proxmox **`pveproxy`**. Proxmox `:8006`, PBS `:8007`, Jellyfin `:8096`, kube API `:6443` are outside top-200 but known-live.

Host	Open ports	Identity
`.1` OPNsense	53,80,443	Unbound DNS + GUI + dual-WAN
`.2` UniFi USW	53,80,443,8080,8443	dnsmasq + UniFi
`.31` Jarvis	22,111,3128,**8000**,9100	SSH · Proxmox · llm_router · exporter
`.50` EX3400	22,830	SSH + NETCONF
`.51-.53` rke2-cp1/2/3	22,6443,9345,10250	RKE2 control plane / etcd
`.100/.199` Ares	21,22,111,139,445,2049,9090	FTP · Samba · NFS · Cockpit
`.148` Homepage	22,8081	SSH · PeaNUT
`.177` Pi-hole 1	22,53,80,443	dnsmasq / pi-hole (primary)
`.178` Pi-hole 2	53,80,443	pi-hole (secondary, DNS HA)
`.179` QuarkyLab	22,111,3128,9100	SSH · Proxmox · exporter
`.180` UPS card	22,80,443	SNMP UPS
`.181` NPM	22,80,81,443	OpenResty proxy
`.182` Vaultwarden	22,80	Vaultwarden
`.183` Observability	22,3000,8080,9100	Grafana · Scrutiny · exporter
`.184` Wazuh	22,443	SIEM dashboard
`.185` Open WebUI	22,8080	Chat UI
`.186` Headscale	22,8080	Mesh VPN control
`.187` Randy	22,111,2049,3128,9100	SSH · NFS · Proxmox · exporter (+8007/8096)
`.201-.204` pve3/4/5/2	22,111,3128,9100	nodes (.204 also 443=step-ca)

Generated 2026-07-13 from live discovery reconciled against the NetFRAME Home-Lab vault.
Sanitized public edition. Public IPs, service tags, serials, and secrets omitted.